

Summary — E-commerce Traffic & Sales Analysis

This project analyzes 349,545 user sessions collected between **2020-11-01 and 2021-01-31**, combining session metadata, account information, and order details. The dataset includes 18 variables describing geography, device, traffic channels, user account status, and product purchases. After cleaning and preprocessing, the analysis focuses on traffic behavior, sales dynamics, customer segmentation, and statistical validation of key hypotheses.

Key Findings

1. Geographic performance

The **Americas** dominate both sales and order volume, driven primarily by the United States, which accounts for over **\$13.9M** in revenue. Asia and Europe follow with significantly lower but stable contributions. The top-selling countries show consistent demand patterns, especially in furniture categories such as sofas, chairs, and beds.

2. Device usage

Desktop users generate **59%** of total revenue, making desktop the primary conversion device. Mobile contributes **39%**, showing stable but lower purchasing power, while tablets account for only **2%**. Sales dynamics confirm that desktop traffic drives the strongest peaks.

3. Traffic channels

Organic Search is the most powerful traffic and sales channel across all devices, followed by Direct and Paid Search. Social Search and Undefined channels contribute minimally. Daily sales patterns show clear weekly seasonality, with peaks on Tuesday–Wednesday and lows on weekends.

4. User account behavior

Only **8%** of sessions belong to registered users, but among them:

- **72%** verified their email
- **17%** unsubscribed from newsletters

Subscribers generate **83%** of total revenue, although unsubscribed users have a slightly higher average order value, indicating occasional high-value purchases.

5. Product categories

The top-selling categories globally and in the United States are identical:

Sofas & armchairs, Chairs, Beds, and Shelving units.

This alignment suggests that the largest market strongly influences global demand.

Statistical Insights

- **Normality tests (Shapiro–Wilk)** show that most traffic channel distributions are non-normal → non-parametric tests required.
- **Kruskal–Wallis test** confirms statistically significant differences in daily session counts across traffic channels.

Two-proportion z-test shows **no significant difference** in the share of organic sessions between Europe and America ($p \approx 0.77$).